IoT Based Monitoring of Foot Pressure Using FSR Sensor

Payal S. Malvade, Atul K. Joshi and Swati P. Madhe

***Abstract*—Analysis of foot plantar pressure distribution is an important factor considered today in the field of healthcare as well as in sports world. Lower body physical rehabilitation therapies are associated with sports world. To avoid long-term damage to the affected part and to accelerate the recovery period, it is necessary to keep an eye on the patient regularly following his/her rehabilitation activities not only in the hospital but also at home. Multiple techniques have been developed for quantitative gait analysis but comparatively these systematic process fails as it measures static or dynamic process for very short duration of time and space. In the proposed system, an electronic insole system is designed in a shoe using force sensing resistor(FSR) sensor that monitors pressure wirelessly, thus saving the information in an array of sensor network for transferring data into CPU. Thus considering the information available used for further study purpose. Four sensors are placed, one at the heel, one at the lateral part, one at the metatarsal head and one at the anterior. The plantar pressure is used to analyse pressure distribution. The signal from the pressure sensor will be processed by microcontroller and sent to mobile via Bluetooth module. With the help of Internet-of-Things(IoT), when the signal goes beyond the limit the notification will be sent to patient consulting the doctors/physiotherapist so that immediate action can be carried out further. The proposed system will help the doctors/physiotherapist as well as person to monitor his health using smart shoe. The device can help the clinician and the patient a tool to monitor and receive condition feedback.**

***Index Terms*—Dynamic position, FSR sensor, In-sole shoe, IoT, Plantar pressure distribution, Rehabilitation, Static position.**

I. INTRODUCTION

IMPROVING the performance in the field of medical management and health maintenance holds an biggest demand in today's era. The need of providing endowment concern to the patient with less medical cost as well as dealing with the availability of nursing staff is a primary issue. Recent development, in the field of Internet-of-things(IoT) is enhancing and improving healthcare as well as biomedical field[1]. -

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Movement examination and determination is the study of human locomotion. These is used to estimate the nature of walk, plan, and treat individuals with conditions affecting their ability to walk. It is mainly needed in monitoring the stepping patterns after any orthopaedic surgical operations during rehabilitation. The difference between the healthy and pathological gait patterns is invisible to the naked eye. As per survey of National institute of ageing, it is being observed that the number is assumed to be triple to about 1.5 billion by 2050, which represents 16 percent of the world's population [2]. With increasing elderly population the health issues like injuries or diseases that cause knee problems, dislocated shoulder are also increasing. So rehabilitation using new systems and strategies after such surgery has became necessary aspect. The improvement and reclamation process plays a pivotal guise in helping the patient to recoup their extremities as well as recapitulate their hasty way of life. The benefits of rehabilitation are relief from pain and swelling, restore normal movement in the joint, build up strength in the joint and surrounding muscles, help with circulation, particularly right after surgery, so that blood clot can be avoided, to prevent enduring caused to the injured spur[3]. So, it is necessary to monitor the patient continuously in clinics as well as in homes[4].

With increase in technology and advancement in the biomedical instrumentation. The new devices has been designed, manufactured, and tested for the monitoring of rehabilitation activities after surgical operations. Before the technological advancement the proper way of an exercise depends only on visual analysis/manual methods. There are number of external measurement system for gait monitoring, which is a important part in rehabilitation procedure. Among them walking aid is most common. If the walking aid pattern is consider then it provides the quantitative data for gait analysis, and these data provides useful information to the orthopaedic so that they can properly instruct the patient's to use it correctly at early stage. The another conventional rehabilitation method i.e proper way of exercise by the patient also depends on visual analysis/ manual methods. This method has disadvantage of human errors such as the perception of the loads on the patient's lower limbs, the movement, the drag, the synchronism, etc. So there is an need of making an in-sole shoe which not only gives quantitative data for gait analysis but also give feedback to the patient for correct use of it. In proposed system the foot plantar pressure distribution is taken into consideration. Hoof load is the stress which operates in the mid of the extremities and surface during daily mobility hustle. During mobility, first step of surface of interaction with

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the environment is provided by feet. In the field of sports and biomedical application "balancing" is a key term. In this the main facts to be considered is pressure distribution on feet which is challenging because pressure of each patient vary according to his/her walking style, age, sex etc as well as patient needs to visit periodically in clinics, so system must be designed which will be portable, low cost, and continuous monitoring with IoT compatibility. Foot plantar pressure measurement can be carried out using two methods i.e platform system and In-shoe system. Platform system are generally embedded in a pathway or track so it can be used only in space or indoor measurements[5]. In-shoe system is embedded in the shoe. Requirement of foot plantar sensors is that they must be very mobile and must have limited cabling [5]. Nowadays in market and research laboratories, number of plantar pressure measurement systems are available. The drawback is that they cannot be used for continuous monitoring or in outside environment. So the device must be made which must have following features. First, it should be easily habiliment to comprehend open-air activities. Secondly, the pressure sensor must have enough correctness. Thirdly, the device must be comfortable and easy to wear for the patient. It is very important to make sure the patient is in a natural status as usual. Fourth, wireless technique must be implemented instead of wired cables[6]. With reference to above four features mentioned, this paper represents an in-sole shoe plantar pressure measurement system in which data acquisition system is designed with small size and stable power supply interference. Bluetooth, a virtual serial port technology is utilized based on which wireless configuration modes is developed to make this system suitable in research laboratories, clinics and daily outdoor/indoor activities. Also real-time monitoring and display of plantar pressure distribution is carried out. The paper is described as follows: The system structure is explained in section II. Section III gives brief introduction of sensor selection and sensor position selection. Section IV gives the concluding remark. Future scope and work to be done is explained in section V.

II. System Structure

The block diagram shown in Fig. 1 consists of:

Pressure Sensor: For monitoring the pressure of the foot, pressure sensor are used. The specific points are mentioned in the relevance(Fig. 3). In the design system the FSR sensor is horsed beneath the heel, the medial, lateral and anterior which is as shown in Fig. 4 [5]. Increase as well the decrease in the pressure will be monitored. Patient physical position is also considered while monitoring.

Signal Conditioning: The pressure sensors are resistive. We need to convert the change in resistance to the change in voltage. The output of the pressure sensor is not sufficient for driving the port of the microcontroller. The output of the sensor must be conditioned for the proper operation purpose.

Microcontroller: The output of a signal conditioning unit is given to the microcontroller. It will collect the data from the sensors and then process it accordingly.

Bluetooth Module: It is communication unit of the system. Bluetooth is used because it is most sophisticated communication system for short distances. Using the Bluetooth communication we will be able to send the data to the receiver.

Mobile Unit: We will develop one app which is dedicated for this system. The application will receive the data from the controller. It will state the condition of the foot based on static and dynamic data.

API Server and Computer: API is application programming interface. The role of API is to secure the data to prevent access from unauthorized person. In this whatever data is there on Mobile through API server it will be sent to computer for further analysis by physiotherapist. This helps the patient as

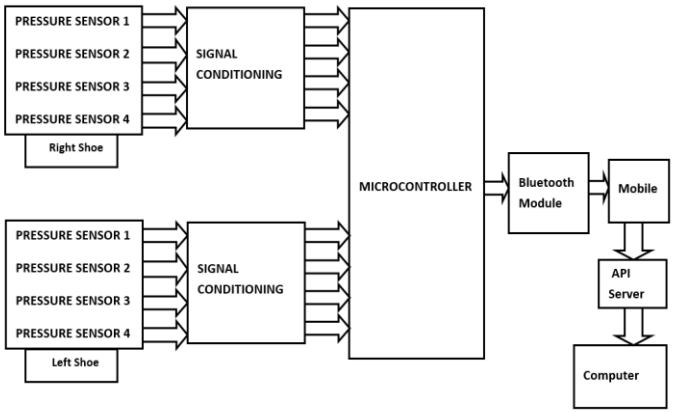

Fig. 1. Block diagram of IoT based in-sole shoe pressure monitoring.

well as physiotherapist to know that how the patient is getting recovered. unit which is an application portal at user mobile.

A. Working of the System

The system works in a simple and sophisticated manner. The first block of a system is pressure sensor. This is the physical block which comes in contact with the patient. The pressure sensors sense the pressure. We are using the resistive pressure sensors. Resistance of the sensor changes as pressure on the observing point increases. Resistive output of a sensor is unable to drive controller. Hence we are using signal conditioning circuit for the conversion of resistive change to the voltage. Once the conversion takes place it is easy for the controller to receive the data. The sensors are now connected to the controller via signal conditioning unit. Controller reads the sensor and stores the value in the system. It checks all the sensor connected one by one and forms an array. This array now needs to be send to the android application through bluetooth. Android application receives the data. If the patient is doing wrong exercise or he/she is taking wrong steps then immediately an alarm will be sent to mobile also that data will be sent to the hospital.

III. Sensor Selection

There are various sensor available for plantar pressure measurement such as (i) piezoelectric pressure sensor, (ii) capacitive, (iii) piezoresistive[7]. Piezoresistive pressure sensors are less susceptibility to noise. Also compared with other sensors this sensor are cost effective. FSR is force

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sensitive resistor whose resistance changes when pressure is applied.

A. Basic Construction and Operation of FSR

The FSR sensor consists of two membranes which are separated by a thin air gap. The rigidity of the two membranes and spacer around the edges are maintained by the air gap. Each membranes has two sets of interdigitated fingers which are electrically separated from each other, where each set connecting to one track on a tail. The other side of membrane is coated with FSR carbon-based ink. The result is that the if

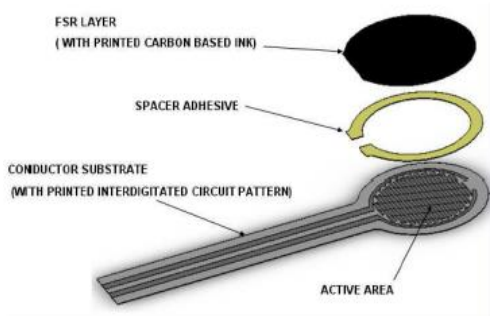

Fig. 2. Basic FSR Construction.

we increase force resistance decreases.

B. Sensor Position Selection

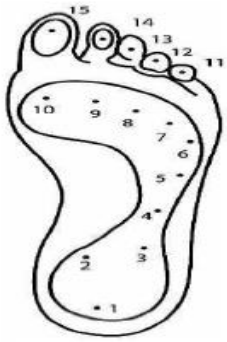

Fig. 3. Foot anatomical areas.

The plantar pressure measured is basically pressure measurement of desired plantar points in the foot. This section involve two decisions one is that to decide the sensor itself and other is to decide the position of the sensor. The foot pressure point is divided into 15 areas, which is shown in Fig. 3. To reduce complexity of the system, five pressure points are considered. Following image shows the points of plantar pressure in Fig.4.

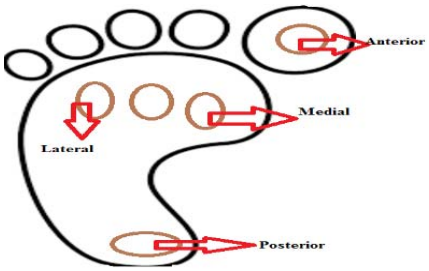

Fig. 4. Selected plantar pressure points.

C. Sensor Characterisation

Sensor characterisation can be carried out by two methods which is universal testing machine and dead-weight pressure gauge. First experimentation is done on universal testing machine. A force was applied on the sensor and corresponding to it graph was observed on computer. Thus observed that as force increases the resistance decreases linearly. As per the relation,

$$\text{Force} = \text{Pressure}/\text{Area} \qquad (1)$$

if we know the force and area, the pressure exerted by each foot point can be known. This quantitative parameters can be helpful for orthopaedicians for gait analysis. Graph of force versus resistance was plotted which is shown in Fig. 6.

Fig. 5. Universal Testing Machine.

TABLE I

READINGS OF UNIVERSAL TESTING MACHINE WITH RESPECT TO RESISTANCE AND FORCE

Sr. No	Force(Newton)	Resistance(Kilo-ohm)
1	75	10.3
2	165	4.7
3	325	3.6
4	560	2.8
5	770	2.5
6	1600	1.9
7	2010	1.3
8	2300	0.8

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Second experimentation was carried out with dead-weight pressure gauge. Experimental set-up of dead-weight pressure gauge shown in Fig. 7. From both the graphs(Fig. 6, Fig. 8) we can conclude that as weight or force increases resistance decreases. The FSR sensor used in the proposed system is more robust but it cannot be used because its response in terms of linearity, and repeatability is poor.

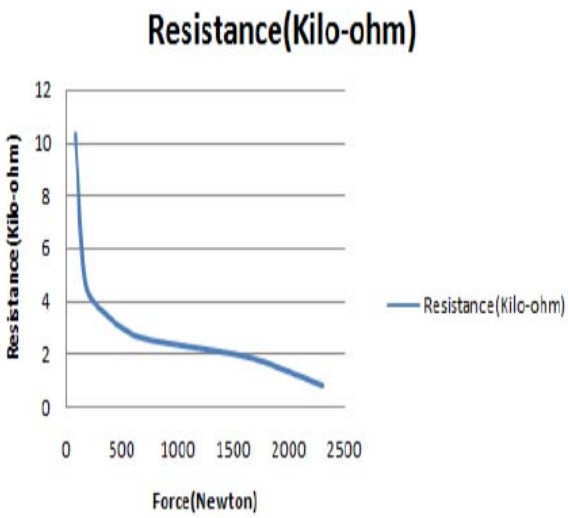

Fig. 6. Graph of resistance versus force.

Fig. 7. Experimental set-up of dead-weight pressure gauge.

TABLE II

READINGS OF DEAD-WEIGHT PRESSURE GAUGE MACHINE WITH RESPECT TO RESISTANCE AND WEIGHT

Sr. No.	Weights(Kg)	Resistance(Kilo-ohm)
1	5.7	13.26
2	8.5	11.52
3	11.3	10.8
4	14.1	8.6
5	16.9	6
6	19.7	5.5
7	22.5	4.5
8	25.3	4.03

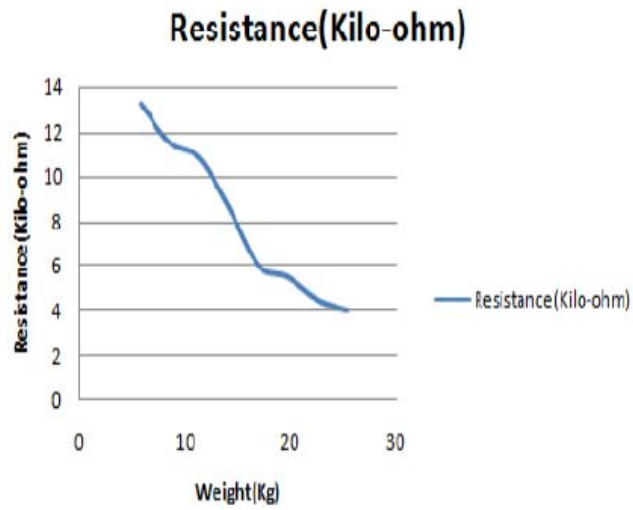

Fig. 8. Graph of resistance versus force.

So next step was using Flexiforce sensor A201. Every layer has a conductive material(silver) applied followed by a layer of conductive ink. Working principle of Flexiforce sensor is same that of FSR sensor. Flexiforce sensor A201 experimentation was carried out with dead-weight pressure gauge. Experimental set-up was similar to Fig. 7.

TABLE III

READINGS OF DEAD-WEIGHT PRESSURE GAUGE MACHINE WITH RESPECT TO RESISTANCE AND WEIGHT

Sr. No.	Weights(Kg)	Resistance(Kilo-ohm)
1	5.7	90.6
2	8.5	71.8
3	11.3	62.3
4	14.1	52.3
5	16.9	45.4
6	19.7	41.8
7	22.5	38.7
8	25.3	35.7

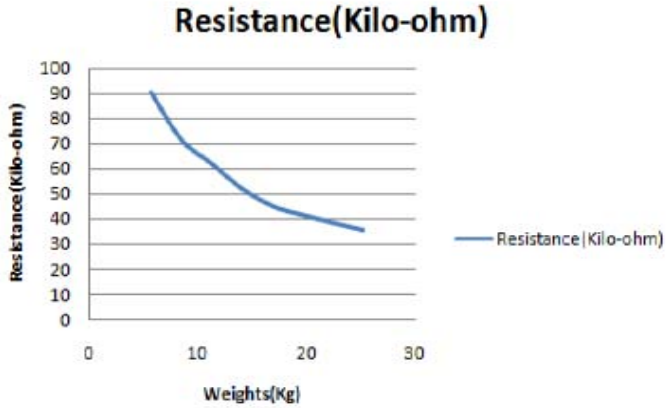

Fig. 9. Graph of resistance versus force.

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Thus from experimentation carried out with FSR sensor and Flexiforce sensor A201 we can conclude that sensor A201 is more linear and repeatable than FSR which is shown in graph(fig. 9). The only drawback of sensor A201 is its cost.

IV. Conclusion

In this system, the insole shoe gives the pressure variation for the patient affected by foot abnormalities, fractures etc. This plantar pressure measurement identifies whether a person is having foot abnormalities or not that will prevent a patient from getting disease. In the system design prevention is taken by the proposed system. The result shows that such a device is reliable and accurately measures the foot pressure distribution to detect abnormalities. Also the size of sensor is very small which is easily mounted on the in-sole. The system is having limited cabling and it is safe to use for the patient.

V. Future Scope

In the future, the proposed system will be added with an Android app that will receive the signal from the transmitter module and alert a healthcare provider if there are abnormalities in the foot pressure distribution. It would also test device on fracture patients and compare the results with a normal foot pressure distribution.

VI. Acknowledgment

The support of the Principal, the Head of the Department (Instrumentation and Control) of MKSSS's Cummins College of Engineering for Women Karvenagar, Pune is gratefully acknowledged. Excellent Guidance of Staff members and Cooperation of non-teaching staff is valuable.

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